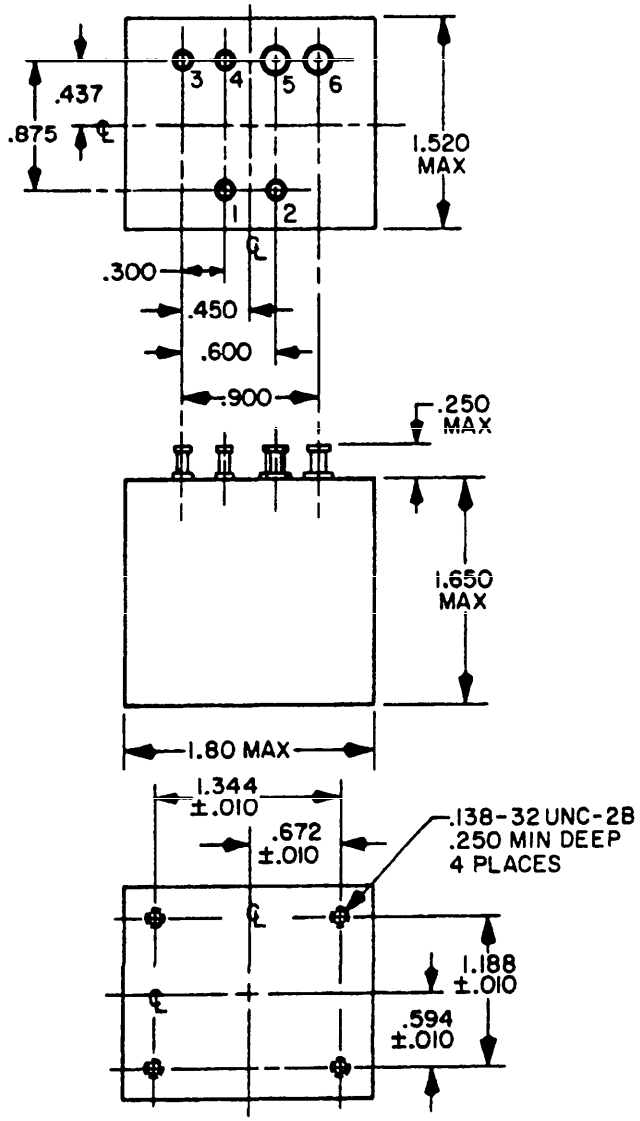
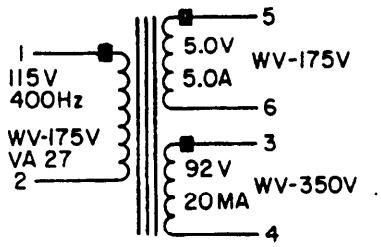
MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-DOWN, TF5V03ZZ919

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformers described herein shall consist of this document and the latest issue of specification MIL-T-27.




CIRCUIT DIAGRAM AND MARKING

| INCHES | MM |
|---|---|
| .010 | .25 |
| .138 | 3.50 |
| .250 | 6.35 |
| .300 | 7.62 |
| .437 | 11.10 |
| .450 | 11.43 |
| .594 | 15.09 |
| .600 | 15.24 |
| .672 | 17.07 |
| .875 | 22.22 |
| .900 | 22.86 |
| 1.188 | 30.18 |
| 1.344 | 34.14 |
| 1.520 | 38.61 |
| 1.650 | 41.91 |
| 1.80 | 45.72 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.030 (.76 mm).
4. Marking shall be on the side of the case.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage:
(1-2) - 115 volts rms, 400 Hz ±5 percent, 27 voltamperes.
Secondary voltage:
(3-4) - 92.0 volts rms, 20 milliamperes rms.
(5-6) - 5.0 volts rms, 5.0 amperes rms.
Working voltage:
(1-2) - 175 volts maximum.
(3-4) - 350 volts maximum.
(5-6) - 175 volts maximum.

Design and construction:

Dimensions and configuration - See figure 1.
Duty cycle - Continuous.
Case - Encapsulated.
Material - Epoxy.
Terminals - Brass, turret type.
Terminal height - 0.250 inch maximum.
Weight - 10.0 ounces maximum.
Operating temperature range - -55° to 155°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 5.0 pounds.

Dielectric withstanding voltage (at sea level):
(1-2) (5-6) - 500 volts rms.
(3-4) - 1000 volts rms.

Electrical characteristics:
No load - With 115 volts rms, 400 Hz across (1-2), current in (1-2) shall not exceed 90 milliamperes rms maximum and the power in (1-2) shall not exceed 2.0 watts.
Voltage across (3-4): 96.77 volts ±2.0 percent.
Voltage across (5-6): 5.26 volts ±2.0 percent.
Rated load-With 115 volts rms, 400 Hz across (1-2), the voltage across (3-4) shall be 92.0 volts rms ±3 percent at 20 milliamperes rms, voltage across (5-6) shall be 5.0 volts rms ±3 percent at 5.0 amperes rms.
Polarity-Additive with terminals 2 and 5, 6 and 3 connected.

Temperature rise: 45°C maximum with 115 volts rms, 400 Hz across (1-2) at an ambient temperature of 110°C.

DC resistance:
(1-2) - 7.0 ohms maximum.
(3-4) - 92.0 ohms maximum.
(5-6) - 0.03 ohm maximum.

Regulation: 6.0 percent maximum.

Marking location: See figure 1.

Part number M27/100 -01.



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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, SG
Navy - EC, SH, OS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - MU, WC, SG, ME
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5950-0511)

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